

Morning Paper Report - Tuesday, August 18, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Synaptic Facilitation: A Key Biological Mechanism for Resource Allocation in Computational Models of Working Memory

Marta Balagué-Marmaña, Laura Dempere-Marco
Cognitive Computation, 2024

Source: Crossref; matched terms: working memory, computational model, computational models, computational models of working memory

Link: <https://doi.org/10.1007/s12559-023-10234-4>

Goal: Abstract Working memory (WM) is a crucial cognitive function required to maintain and manipulate information that is no longer present through the senses

Method: A biophysically inspired attractor model of WM, endowed with synaptic facilitation, is considered in this study

Results: Although the network connectivity determines the WM capacity by regulating the excitation-inhibition balance, the display protocol modulates its effective limit Our findings shed light on how different stimulation protocol dynamics affect WM, underscoring the importance of synaptic facilitation and experimental protocol design in modulating WM capacity.

Conclusion: Although the network connectivity determines the WM capacity by regulating the excitation-inhibition balance, the display protocol modulates its effective limit Our findings shed light on how different stimulation protocol dynamics affect WM, underscoring the importance of synaptic facilitation and experimental protocol design in modulating WM capacity.

2. Artificial Intelligence Software Structured to Simulate Human Working Memory, Mental Imagery, and Mental Continuity

J. Reser
arXiv.org, 2022

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks, artificial intelligence

Link: <https://www.semanticscholar.org/paper/353891dd5038780069ebf0c1c0ec777cb2632d14>

Goal: This article presents an artificial intelligence (AI) architecture intended to simulate the iterative updating of the human working memory system

Method: It features several interconnected neural networks designed to emulate the specialized modules of the cerebral cortex These are structured hierarchically and integrated into a global workspace

Results: This maintenance is made possible by persistent neural activity in the form of two modalities: sustained neural firing (resulting in a focus of attention) and synaptic potentiation (resulting in a short-term store)

Conclusion: The present article will explore how this architecture can lead to iterative shift in the distribution of coactive representations, ultimately leading to mental continuity between processing states, and thus to human-like thought and cognition Taken together, these components outline a biologically motivated route toward synthetic consciousness or artificial sentience and subjectivity.

3. Corrigendum: Alterations in Neural Networks During Working Memory Encoding Related to Cognitive Impairment in Temporal Lobe Epilepsy

Liping Pan, Yakun Wu, J. Bao, Dandan Guo, Xin Zhang, Jiajing Wang, et al.
Frontiers in Human Neuroscience, 2022

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/8ba659c2a47e2592045e6781d3fe571550bfa585>

Goal: [This corrects the article DOI: 10.3389/fnhum.2021.770678.].

Method: [This corrects the article DOI: 10.3389/fnhum.2021.770678.].

Results: [This corrects the article DOI: 10.3389/fnhum.2021.770678.].

Conclusion: [This corrects the article DOI: 10.3389/fnhum.2021.770678.].

4. Pattern memory cannot be completely and truly realized in deep neural networks

Tingting Li, Ruimin Lyu, Zhenping Xie
Scientific Reports, 2024

Source: Semantic Scholar; matched terms: neural network, neural networks

Link: <https://www.semanticscholar.org/paper/73c807c1585a884c01660ab63fcc78af121a6823>

Goal: The unknown boundary issue, between superior computational capability of deep neural networks (DNNs) and human cognitive ability, has becoming crucial and foundational theoretical problem in AI evolution

Method: Inspired by perceptual characteristics of human vision on optical illusions, we propose a novel working capability analysis framework for DNNs through innovative cognitive response characteristics on visual illusion images, accompanied with fine adjustable sample image construction strategy

Results: The unknown boundary issue, between superior computational capability of deep neural networks (DNNs) and human cognitive ability, has becoming crucial and foundational theoretical problem in AI evolution

Conclusion: Our findings indicate that, although DNNs can infinitely approximate human-provided empirical standards in pattern classification, object detection and semantic segmentation, they are still unable to truly realize independent pattern memorization

5. Effects of indoor exposure to low level toluene on neural network alterations during working memory encoding.

Liping Pan, Jie Liu, Changqing Zhan, Xin Zhang, Mingrui Cui, Xiao Su, et al.
Chemosphere, 2023

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/616cddaaf8f74b12c8c4e7b75f1d55f0eccfcb6b>

Goal: OBJECTIVE While high concentrations of toluene are known to affect multiple human organ systems, research concerning the influence of immediate, short-term exposure to toluene indoors and at low concentrations is scarce

Method: All participants were situated in a closed environmental chamber with a full fresh air system

Results: RESULTS

Conclusion: The DTFFZ-F7 values of Group D were higher as compared to those of the Group C CONCLUSION Short-term toluene exposure significantly influences neural networks during cognitive processes such as WM encoding, even at low concentration.